

Chapter 4 Maxwell's Equations

1. Electromagnetic Induction

Consider two long, parallel wires that carry equal positive charge but are stationary in the lab frame. In this frame, the wires repel each other purely due to the electrostatic (Coulomb) force. However, when observed from a moving reference frame that travels parallel to the wires at a constant velocity, the charges now appear to be in motion and thus form steady currents. In this frame, each wire generates a magnetic field that interacts with the current in the other wire, resulting in a magnetic force between them.

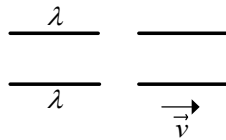


Figure. The force between two charge-carrying wires is purely electric in the rest frame (left).

When viewed from a moving frame, the wires carry steady currents, and a magnetic force arises between them (right).

This leads to a deeper physical insight: electric and magnetic forces are manifestations of the same underlying phenomenon—electromagnetism—and are frame-dependent. What appears purely electric in one frame may involve magnetic effects in another, illustrating how special relativity and electromagnetism are intrinsically connected. The transformation of electric fields into magnetic fields under a change of inertial frames shows that magnetism can be interpreted as a relativistic correction to electrostatics.

1) The Maxwell—Faraday Equation

In the early 19th century, Michael Faraday made a groundbreaking discovery: electromagnetic induction. While Oersted had earlier demonstrated that electric currents can generate magnetic fields, Faraday posed a bold and revolutionary question—could magnetic fields, in turn, generate electric currents? His experiments confirmed that a changing magnetic field can induce an electric current in a nearby conductor, unveiling the phenomenon of induction. This insight revealed that electricity and magnetism are not just related—they are dynamically intertwined. Faraday's discovery laid the foundation for Maxwell's equations, the unifying framework of classical electromagnetism, and

sparked the technological revolution that led to electric generators, transformers, and the modern power grid. Below, I will demonstrate how the Maxwell-Faraday equation—the fundamental expression governing electromagnetic induction—can be derived from electrostatics and the principles of special relativity. Consider the following situation: a charge distribution $\rho_0(x', y', z')$ is static in the O' frame, which moves at a constant velocity $\vec{u}$ along the x' direction relative to the O frame, as shown in the figure below.

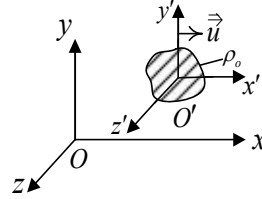


Figure. A distribution of charge is static in the O' frame, which moves at a constant velocity relative to the O frame.

In the O' frame, the charge distribution is static, so electrostatic laws apply within the x' - y' - z' coordinate system:

$$\nabla' \times \vec{E}' = \partial_{x'} E_x' + \partial_{y'} E_y' + \partial_{z'} E_z' = \rho_0(x', y', z')/\epsilon_0,$$

$$\nabla' \times \vec{E}' = 0,$$

Or, written in the component form,

$$\begin{cases} \partial_{y'} E_z' - \partial_{z'} E_y' = 0 \\ \partial_{z'} E_x' - \partial_{x'} E_z' = 0. \\ \partial_{x'} E_y' - \partial_{y'} E_x' = 0 \end{cases}$$

Apparently, there is no magnetic field in the O' frame. However, according to special relativity, a magnetic field is generated in the O frame (we have not learnt special relativity yet, for details, please refer to Griffiths' book, Equation 12.109 on page 559):

$$\begin{cases} E_x = E_x' \\ E_y = \gamma E_y' \\ E_z = \gamma E_z' \\ B_x = 0 \\ B_y = \frac{\gamma u}{c^2} E_z' \\ B_z = \frac{\gamma u}{c^2} E_y' \end{cases},$$

where $\gamma = \frac{1}{\sqrt{1-u^2/c^2}}$ is the Lorentz factor. Meanwhile, the partial derivatives undergo transformations as:

$$\begin{cases} \partial_{x'} = \gamma \left(\partial_x + \frac{u}{c^2} \partial_t \right) \\ \partial_{y'} = \partial_y, \partial_{z'} = \partial_z \\ \partial_{t'} = \gamma (\partial_t + u \partial_x) \end{cases}$$

Substituting the above results into $\nabla' \times \vec{E}' = 0$ and transforming to the unprimed coordinates, we obtain:

$$\begin{cases} \partial_{y'} E_{z'}' - \partial_{z'} E_{y'}' = \frac{1}{\gamma} (\partial_y E_z - \partial_z E_y) = 0 \\ \partial_{z'} E_{x'}' - \partial_{x'} E_{z'}' \\ \quad = \partial_z E_x - \gamma \left(\partial_x + \frac{u}{c^2} \partial_t \right) \frac{E_z}{\gamma} \\ \quad = \partial_z E_x - \partial_x E_z - \frac{u}{c^2} \partial_t E_z \\ \quad = \partial_z E_x - \partial_x E_z + \partial_t B_y = 0 \\ \partial_{x'} E_{y'}' - \partial_{y'} E_{x'}' = \partial_x E_y - \partial_z E_x + \partial_t B_z = 0 \end{cases}$$

In sum, the above simple example demonstrates that the following equation holds in the O frame:

$$\boxed{\nabla \times \vec{E} = -\frac{\partial \vec{B}}{\partial t}}$$

This is the well-known Maxwell–Faraday equation, one of Maxwell’s equations in differential form. It succinctly expresses the fundamental coupling between electric and magnetic fields. It is remarkable that the spatial variation of the electric field is directly related to the negative time rate of change of the magnetic field.

The similarity between the Maxwell-Faraday equation and Ampere’s Law ($\nabla \times \vec{B} = \mu_0 \vec{J}$) allow us to visualize how a time-changing magnetic field induces an electric field, as shown in the following figure.

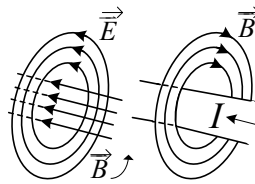


Figure. Parallelism between Faraday’s Law (left) and Ampere’s Law (right).

The Maxwell-Faraday equation reveals that the electric field is no longer irrotational in the presence of a time-varying magnetic field. So it is justifiable to decompose any electric field into two parts:

① Irrotational part ($\vec{E}_{ir}$)

$\nabla \times \vec{E}_{ir} = 0$, $\vec{E}_{ir} = -\nabla \phi$, where ϕ is the electric scalar potential.

② Solenoidal part ($\vec{E}_{so}$)

$\nabla \times \vec{E}_{so} = -\partial_t \vec{B}$. Since $\vec{B} = \nabla \times \vec{A}$, so $\nabla \times \vec{E}_{so} = -\partial_t (\nabla \times \vec{A}) = \nabla \times (-\partial_t \vec{A})$. Accordingly,

$$\vec{E}_{so} = -\partial_t \vec{A}.$$

In sum, the total electric field, $\vec{E}_{tot}$, can be written as:

$$\boxed{\vec{E}_{tot} = \vec{E}_{ir} + \vec{E}_{so} = -\nabla\phi - \partial_t \vec{A}}.$$

2) Faraday's Law of Induction

Integrating both sides of the Maxwell-Faraday equation obtained above over an open surface:

$$\int_{S(t)} \nabla \times \vec{E} \cdot d\vec{a} = - \int_{S(t)} \frac{\partial \vec{B}}{\partial t} \cdot d\vec{a}$$

Note that both $\vec{E}$ and $\vec{B}$ depend on space and time, and here we also allow the open surface to vary with time. The left-hand side of the equation can be expressed as a line integral around the boundary ($P(t)$, which is also time dependent) Stokes' Theorem:

$$\oint_{P(t)} \vec{E} \cdot d\vec{\ell} = - \int_{S(t)} \frac{\partial \vec{B}}{\partial t} \cdot d\vec{a}.$$

The line integral of the electric field has units of voltage (or potential) and is traditionally referred to as the electromotive force (EMF), even though it is not a force. Just as in electrostatics, this potential can do work on charges. If a real conducting wire is placed along the loop, as illustrated in the following figure, the induced potential drives the motion of charges around the loop.

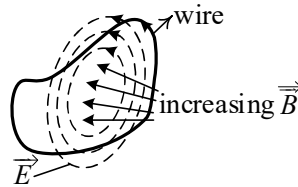


Figure. Placing a conducting wire along the loop.

An interesting scenario arises when the magnetic field remains fixed while the wire moves. To fully account for the time dependence of both $\vec{B}$ and the evolving surface $S(t)$ (and its boundary $P(t)$), we invoke the Reynolds transport theorem (a generalization of the Leibniz integral rule):

$$\frac{d}{dt} \int_{S(t)} \vec{B} \cdot d\vec{a} = \int_{S(t)} \frac{\partial \vec{B}}{\partial t} \cdot d\vec{a} - \oint_{P(t)} (\vec{v} \times \vec{B}) \cdot d\vec{\ell}$$

where $\vec{v}$ specifies the spatially-dependent velocity of the boundary. Combining the above two equations, we get:

$$\oint_{P(t)} (\vec{E} + \vec{v} \times \vec{B}) \cdot d\vec{\ell} = -\frac{d}{dt} \int_{S(t)} \vec{B} \cdot d\vec{a} = -\frac{d}{dt} \Phi_B$$

In the expression above, the left-hand side consists of two components. The first term, $\oint_P \vec{E} \cdot d\vec{\ell}$, is known as the transformer EMF, which arises from a time varying magnetic field. The second term, $\oint_P (\vec{v} \times \vec{B}) \cdot d\vec{\ell}$, termed the motional EMF, results from the motion of the conducting wire through a magnetic field. In the case of a stationary loop with a time-varying magnetic field, only the transformer EMF is present, and the equation reduces to:

$$\oint_P \vec{E} \cdot d\vec{\ell} = -\frac{d}{dt} \int_S \vec{B} \cdot d\vec{a}$$

This is the Faraday's Law of induction, one of Maxwell's equations in integral form.

3) Transformer EMF and Motional EMF

In the above derivation, we obtain the term

$$\oint_{P(t)} (\vec{v} \times \vec{B}) \cdot d\vec{\ell},$$

which represents the motional EMF from a purely mathematical perspective. The following analysis offers deeper physical insight into this result. Assume the magnetic field is stationary while the wire is moving, from time t_0 to $t_0 + \delta t$, the path evolves from $P(t_0)$ to $P(t_0 + \delta t)$, as shown in the following figure.

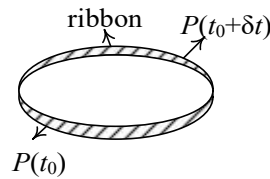


Figure. Illustration of a time-varying wire.

During the short time interval δt , the wire sweeps out a surface that forms a ribbon-like strip. If we denote the surfaces bounded by the wire at times t_0 and $t_0 + \delta t$ as $S(t_0)$ and $S(t_0 + \delta t)$, respectively, then together with the ribbon, they enclose a thin, disk-like volume. Integrating $\vec{B}$ over the surface of this thin disk, we obtain:

$$\int_{\text{disk}} \vec{B} \cdot d\vec{a} = - \int_{S(t_0)} \vec{B} \cdot d\vec{a} + \int_{\text{ribbon}} \vec{B} \cdot d\vec{a} + \int_{S(t_0 + \delta t)} \vec{B} \cdot d\vec{a} = 0$$

The minus sign in front of $\int_{S(t_0)} \vec{B} \cdot d\vec{a}$ accounts for the oppositely-pointing surface normal. The integral vanishes because of the absence of magnetic monopole. The integral over the top and bottom surfaces are related to the magnetic flux at $t_0 + \delta t$ and t_0 :

$$\begin{aligned} \int_{S(t_0+\delta t)} \vec{B} \cdot d\vec{a} &= \Phi_B(t_0 + \delta t), \\ - \int_{S(t_0)} \vec{B} \cdot d\vec{a} &= -\Phi_B(t_0). \end{aligned}$$

We now focus on the integral over the ribbon region by examining a small segment of it, as illustrated in the figure below.

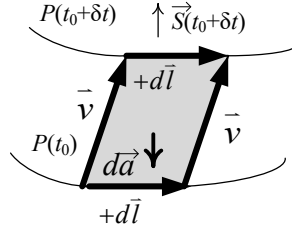


Figure. A zoomed-in view of a small segment of the ribbon.

Note the indicated positive directions shown in the figure: $\vec{S}(t_0 + \delta t)$ and $d\vec{\ell}$ must satisfy the right-hand rule. On the ribbon, the surface element $d\vec{a}$ (shown as the shaded region) points outward, which can be expressed as:

$$d\vec{a} = d\vec{\ell} \times \vec{v} dt,$$

accordingly, the integral $\int_{\text{ribbon}} \vec{B} \cdot d\vec{a}$ can be evaluated to be:

$$\int_{\text{ribbon}} \vec{B} \cdot d\vec{a} = \oint_{P(t_0)} \vec{B} \cdot (d\vec{\ell} \times \vec{v} dt) = -dt \oint_{P(t_0)} \vec{B} \cdot (\vec{v} \times d\vec{\ell}).$$

This result, when substituted into $-\int_{S(t_0)} \vec{B} \cdot d\vec{a} + \int_{\text{ribbon}} \vec{B} \cdot d\vec{a} + \int_{S(t_0+\delta t)} \vec{B} \cdot d\vec{a} = 0$, yields

$$\Phi_B(t_0 + \delta t) - \Phi_B(t_0) = \left(\oint_{P(t_0)} \vec{B} \cdot (\vec{v} \times d\vec{\ell}) \right) dt.$$

From the above equation, we readily recover the contribution of the motional EMF to the rate of change of magnetic flux, consistent with our earlier result:

$$\oint_{P(t)} (\vec{v} \times \vec{B}) \cdot d\vec{\ell} = - \frac{d\Phi_B}{dt}$$

Inside the wire, charges flow at a velocity $\vec{u}$ relative to the wire, while the wire itself moves at a velocity $\vec{v}$. AS a result, the total velocity of the charges is $\vec{w} = \vec{u} + \vec{v}$. We can show that $\vec{w} \times d\vec{\ell} = \vec{u} \times d\vec{\ell} + \vec{v} \times d\vec{\ell} = \vec{v} \times d\vec{\ell}$, since $\vec{u} \times d\vec{\ell} = 0$ (they are parallel). Accordingly,

$$\begin{aligned} \oint_{P(t_0)} \vec{B} \cdot (\vec{v} \times d\vec{\ell}) &= \oint_{P(t_0)} \vec{B} \cdot (\vec{w} \times d\vec{\ell}) = - \oint_{P(t_0)} (\vec{w} \times \vec{B}) \cdot d\vec{\ell}, \\ \oint_{P(t)} (\vec{w} \times \vec{B}) \cdot d\vec{\ell} &= - \frac{d\Phi_B}{dt} \end{aligned}$$

The meaning of the motional EMF is now clear: $\vec{w} \times \vec{B}$ represents the Lorentz force per unit charge, and $\oint_{P(t)} (\vec{w} \times \vec{B}) \cdot d\vec{\ell}$ is this force integrated along the loop, which is equivalent to a potential. Note that one might think the magnetic force is always perpendicular to the motion of the charges, and therefore the integral should vanish. However, this is incorrect, because the magnetic force is not necessarily perpendicular to the loop element $d\vec{\ell}$, along which the integration is performed. To conclude, we denote the total EMF as $\mathcal{E}$ and the magnetic force per unit charge as $\vec{f}_B$:

$$\mathcal{E} = - \frac{d\Phi_B}{dt} = \underbrace{\oint_{P(t)} \vec{E} \cdot d\vec{\ell}}_{\text{transformer EMF}} + \underbrace{\oint_{P(t)} \vec{f}_B \cdot d\vec{\ell}}_{\text{motional EMF}}$$

Note:

- ① A simple setup that illustrates how magnetic forces generate motional EMF

A conducting bar slides along two metal rails, with a magnetic field present between them (see the figure below).

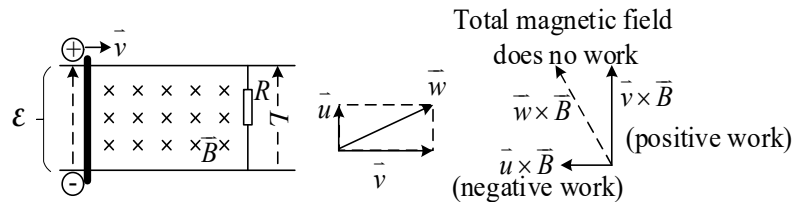


Figure. A conducting bar in motion generates motional EMF (left). Velocity decomposition of the charge (middle). Force decomposition (right).

As the bar moves through the magnetic field, it cuts across magnetic field lines, causing positive charges to be pushed upward and negative charges downward, thereby establishing a voltage across

the bar. The overall motion of the charges—and the magnetic force acting on them—can be decomposed into two components. Although the total magnetic force does no net work on the charges, its vertical component, which gives rise to the motional EMF, consistently contributes positively. While transformer EMF converts energy associated with a time-varying magnetic field into electric energy, motional EMF converts mechanical (kinetic) energy into electric energy.

② There is sometimes no clear-cut distinction between transformer EMF and motional EMF

Refer to the figure below. If the wire moves toward the magnet (left), the induced current is attributed to motional EMF. Conversely, if the wire remains stationary while the magnet moves in the opposite direction (right), the current arises from transformer EMF. However, since motion is relative, whether the EMF is motional or transformer depends on the chosen reference frame.

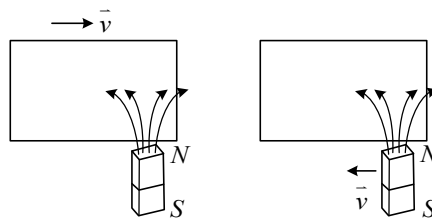


Figure. Motional EMF (left) and transformer EMF (right).

③ Explain the minus sign in Faraday's Law of Induction

To demonstrate the importance of the minus sign, consider an example where the north pole of a magnet moves toward a conducting ring.

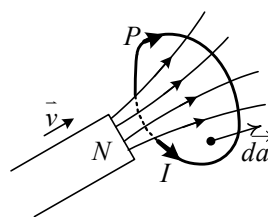


Figure. The north pole of a magnet is approaching a conductor ring.

We define a positive direction for $d\vec{a}$ (here, to the right), which also determines the positive direction of P (clockwise). In this convention, $\frac{d\Phi_B}{dt}$ is positive so $\mathcal{E} = -\frac{d\Phi_B}{dt}$ is negative, indicating a counter-clockwise induced current. This current generates its own magnetic field, opposing the change in magnetic flux—a form of negative feedback that prevents runaway energy buildup. This fundamental principle is known as Lenz's Law.